

Here are Python questions covering List, Tuple, Set, Dictionary, and String Manipulation.

1. Employee Attendance System (List)

Question:

A company stores employee IDs who attended a training session in a list. Find the total number of employees and remove duplicate entries.

Sample Input:

```
employees = [101, 102, 103, 101, 104, 102, 105]
```

Sample Output:

Total Employees: 7

Unique Employees: [101, 102, 103, 104, 105]

2. Online Shopping Cart (List)

Question:

An e-commerce website stores products added to a cart in a list. Remove an item that the customer canceled and add a new product.

Sample Input:

```
cart = ["Laptop", "Mouse", "Keyboard", "Headphone"]
```

```
cancel = "Mouse"
```

```
new_item = "Webcam"
```

Sample Output:

Updated Cart:

```
['Laptop', 'Keyboard', 'Headphone', 'Webcam']
```

3. Student Marks Record (Tuple)

Question:

A student's marks are stored in a tuple. Find the highest and lowest marks.

Sample Input:

```
marks = (78, 90, 67, 88, 95, 72)
```

Sample Output:

Highest Mark: 95

Lowest Mark: 67

4. Product Price Details (Tuple)

Question:

A tuple contains product information:
(Product ID, Product Name, Price).

Display the product name and price.

Sample Input:

```
product = (101, "Laptop", 55000)
```

Sample Output:

Product Name: Laptop

Price: 55000

5. Company Email Registration (Set)

Question:

A company wants to store unique employee email IDs. Remove duplicate email addresses.

Sample Input:

```
emails = {  
    "john@gmail.com",  
    "sam@gmail.com",  
    "john@gmail.com",  
    "alex@gmail.com"  
}
```

Sample Output:

Unique Emails:

```
{'john@gmail.com', 'sam@gmail.com', 'alex@gmail.com'}
```

6. Social Media Followers (Set)

Question:

Two Instagram users have followers stored in sets. Find mutual followers.

Sample Input:

```
user1 = {"A", "B", "C", "D"}
```

```
user2 = {"C", "D", "E", "F"}
```

Sample Output:

Mutual Followers:

```
{'C', 'D'}
```

7. Employee Salary Management (Dictionary)**Question:**

A company stores employee salaries in a dictionary. Increase each salary by 10%.

Sample Input:

```
salary = {  
    "John": 30000,  
    "Sam": 40000,  
    "Alex": 50000  
}
```

Sample Output:

```
{  
    'John': 33000,  
    'Sam': 44000,  
    'Alex': 55000  
}
```

8. Student Result System (Dictionary)**Question:**

Store student marks in a dictionary and display the student with the highest mark.

Sample Input:

```
marks = {  
    "Ravi": 78,  
    "Priya": 92,  
    "Kumar": 85,  
    "Anu": 96  
}
```

Sample Output:

Topper: Anu

Marks: 96

9. Email Validation (String Manipulation)**Question:**

Check whether an email contains '@' and '.com'.

Sample Input:

email = "student@gmail.com"

Sample Output:

Valid Email

Sample Input:

email = "studentgmail.com"

Sample Output:

Invalid Email

10. Customer Review Analysis (String Manipulation)**Question:**

Count the number of words in a customer review.

Sample Input:

review = "The product quality is very good and delivery was fast"

Sample Output:

Total Words: 10

Combination of All**Question:**

An online store has:

products = ["Laptop", "Mouse", "Laptop", "Keyboard"]

prices = {

 "Laptop": 55000,

```
"Mouse": 500,  
"Keyboard": 1200  
}
```

1. Remove duplicate products.
2. Display each product with its price.
3. Find the total bill.

Sample Output:

Laptop : 55000

Mouse : 500

Keyboard : 1200

Total Bill: 56700

These questions closely resemble **real-world scenarios asked in Python interviews and coding rounds.**